

FRESHVII

A food lifecycle and rescue system.

What should I cook today so this food doesn't become tomorrow's waste?

FRESHNESS STATES

- Fresh
- Use Soon
- Rescue Today
- Expired

THE PROBLEM

People don't mean to waste food. They forget what they already have.

01

Forget what's inside the fridge

02

Forget when food was opened

03

Lose track of partly used ingredients

04

Freeze food, still get irrelevant expiry alerts

05

Struggle to decide what to cook first

06

Buy groceries while older food sits unused

WHY LABELS FALL SHORT

Expiration labels are static. Food isn't.

Opened

A sealed item starts a new, shorter clock.

Frozen

Moving to the freezer slows spoilage.

Partly used

Some of the ingredient stays behind.

Cooked

Cooking creates leftovers with their own timer.

Moved

Fridge, freezer and pantry age food differently.

FRESHVII tracks the **lifecycle of stored food, not just the printed date.**

TARGET USER

Anyone who keeps food in a fridge, freezer or pantry

Primary users

Students

Families

Working individuals

Meal-preppers

Budget-conscious households

CORE USER NEED

“Tell me what food I should use next before it becomes waste.”

THE SOLUTION

A smart digital refrigerator, on your phone

FRESHVII is a mobile-first Progressive Web App that helps you:



Track stored food



Estimate freshness



**Update food as its
condition changes**



**Spot ingredients at risk
of spoiling**



**Recommend meals that use
them first**



**Track food consumed,
preserved or discarded**

CORE VALUES

Four values shape every screen

01

Visibility

Know what you have.

A visual fridge with Fridge, Freezer and Pantry tabs. Status shown by label, icon and color, never color alone.

02

Awareness

Know what needs attention.

An estimated freshness state and Rescue Score for every item. Never a fake exact spoilage date.

03

Action

Know what to do next.

Rescue My Food turns warnings into meals that use at-risk ingredients first.

04

Adaptability

Food changes. So does the tracker.

Opened, frozen, partly used, cooked: every change recalculates freshness.

AWARENESS

A Rescue Score ranks what to use first

Food	Status	Rescue Score
Chicken	Rescue Today	95
Tomatoes	Use Soon	82
Milk	Use Soon	61
Eggs	Fresh	20

Chicken Breast

Freshness: 27%



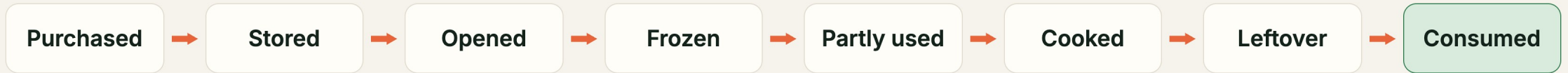
Rescue Today

Estimated remaining freshness: about 1 day

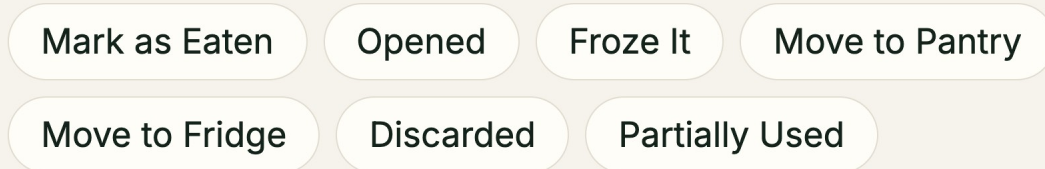
Estimates use food category, date stored, typical shelf life, opened or not, storage location and frozen status. FRESHVII shows an estimated state, never an exact spoilage date.

ADAPTABILITY

Food changes. The tracker changes with it.



One-tap quick actions



WHAT THIS PREVENTS

You freeze the chicken, but the app still says it expires tomorrow.

Every state change recalculates freshness.

MVP USER FLOW

Six steps from grocery bag to empty plate

1

Add food

Camera, barcode, photo or manual. The user confirms before saving.

2

Choose storage

Fridge, freezer or pantry, plus quantity, unit, date and opened status.

3

Visual fridge

The item appears in the digital fridge with its freshness state.

4

State changes

Consume, open, freeze, discard or update quantity with a swipe or tap.

5

Rescue My Food

High-risk ingredients drive meal recommendations.

6

Cook

Log how much was used. Leftovers become new items with a fresh timer.

MVP SCREENS

Only six screens for the hackathon

1 • Home / Visual Fridge

Fridge, Freezer, Pantry tabs; visual fridge; freshness indicators; at-risk count; Rescue My Food button.

2 • Add Food

Camera, barcode or manual; name, quantity, unit, storage, opened status; confirmation.

3 • Food Details

Quantity, freshness, time left, storage, opened status. Consumed, Opened, Freeze, Discard, Edit.

4 • Rescue My Food

Urgent ingredients, Rescue Scores, recommended meals, number of ingredients rescued.

5 • Recipe

Recipe, ingredients on hand and missing, prep time, why it's recommended, Cook This.

6 • Consumption Confirmation

How much did you use? Any leftovers? Inventory updates.

DEMO SCENARIO

The demo fridge, prepared in advance

FridgeFreezerPantry

Chicken breast
500 g · Opened

Rescue Today

Leftover rice
400 g

Rescue Today

Tomatoes
3 pieces

Use Soon

Milk
700 ml · Opened

Use Soon

Beef
500 g · Freezer

Fresh

AT RISK RIGHT NOW

2

Chicken and leftover rice need rescuing today.

Rescue My Food

DEMO SCRIPT

One lifecycle, eight beats

1

Open FRESHVII. "This is my digital fridge."

2

Chicken and rice show as at risk.

3

Tap **Rescue My Food**.

4

Get **Chicken Fried Rice**: chicken, leftover rice, optional tomatoes.

5

Tap **Cook This**.

6

"Did you use all 500 g of chicken?" Select 300 g.

7

200 g remains. Mark it frozen; freshness recalculates.

8

Show impact: **3 ingredients rescued**.

IMPACT

3 ingredients rescued.

Show food rescued, items not discarded and money saved. Honest, simple numbers beat unrealistically precise environmental claims.

TECHNOLOGY

The innovation is the combination, not the camera

Frontend stack

React

TypeScript

Vite

Tailwind CSS

shadcn/ui

PWA

Camera APIs

TanStack Query?

Frontend runs on mock data while backend defines the DTOs and food-state structure.

What's innovative

 **Dynamic freshness estimates**

 **Food lifecycle state changes**

 **Rescue Score**

 **Risk-prioritized recipe recommendations**

 **PWA with mobile camera support**

JUDGING STRATEGY

Designed around how we'll be scored

Category	Weight	How FRESHVII earns it
Product · Relevance	5%	Food waste is obvious in the first seconds of the demo
Product · Impact & value	10%	Items rescued and money saved, without fake precision
Product · UI/UX	10%	The visual fridge is understandable before we explain it
Product · Sustainability	10%	Waste reduction built into core behavior, not a closing slide
Tech · Functionality	15%	Add → Store → Track → Rescue → Cook → Update works end to end
Tech · Innovation	15%	Freshness estimates, lifecycle states, Rescue Score, risk-ranked recipes
Creativity · Originality	10%	A food lifecycle and rescue system, not an expiration tracker
Creativity · Innovation	10%	Interactive fridge and Rescue My Food: what needs saving today?
Presentation	15%*	Clear story plus a demo that shows the whole lifecycle

FRESHVII · 15 *Rubric lists 10% clarity + 10% demo under a 15% heading. Verify the official rubric.

PITCH STRUCTURE

The story in six beats

1

Problem

People don't intend to waste food. They forget what they have.

2

Limitation

Expiration dates are static, but food isn't.

3

Solution

FRESHVII follows food as it's opened, frozen, eaten, cooked and stored again.

4

Demo

Show the lifecycle.

5

Impact

Show what was rescued.

6

Future

Sensors, smarter predictions, grocery integrations.

SCOPE

A smaller working product beats ten half-built features

Building now

The core loop: **Add → Store → Track → Rescue → Cook → Update inventory**

Only if the MVP is stable: proactive notifications, a weekly pre-grocery recap, dietary preference filters.

Roadmap, not now

Social features

Gamification

Advanced accounts

Grocery ordering

Complex AI chat

IoT fridge hardware

Expensive sensors

Massive recipe databases

Environmental calculations

Complicated analytics

Elaborate animations

TEAM RESPONSIBILITIES

Five roles, one working demo

Leader / Pitcher

[Name]

Product scope and decisions, MVP, coordination, judging alignment, demo scenario, pitch and storytelling.

UI/UX

[Name]

User flow, visual fridge, wireframes, interaction design, hierarchy, states, frontend handoff.

Frontend

[Name]

PWA, screens, interactions, camera integration, reusable UI, backend integration, responsive behavior.

Backend

[Name]

DTOs, food model, state transitions, freshness logic, Rescue Score, recipe APIs, persistence.

QA / Product Critic

[Name]

Product flaws, edge cases, test scenarios, assumptions, usability review, demo risks, judge questions.

DEFINITION OF DONE

A feature is finished only when all seven are true

✓ The UI exists

✓ The interaction works

✓ The backend supports it

✓ Data updates properly

✓ QA has tested it

✓ The demo can reproduce it

✗ We can explain why it matters

BEYOND THE HACKATHON

What needs to be saved today?

Sensors

Optional temperature data to sharpen estimates.

Smarter predictions

Freshness models that learn from real use.

Grocery integrations

Use what you have before you buy more.